

Wingtip vortices

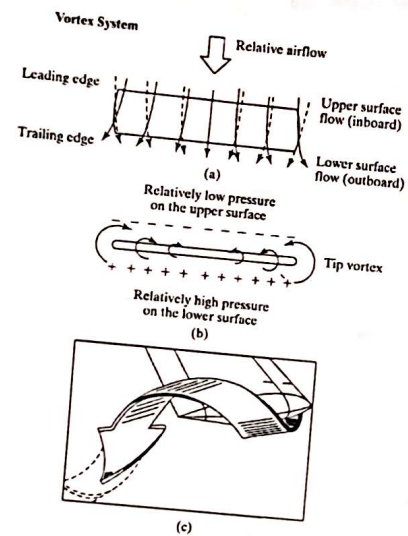
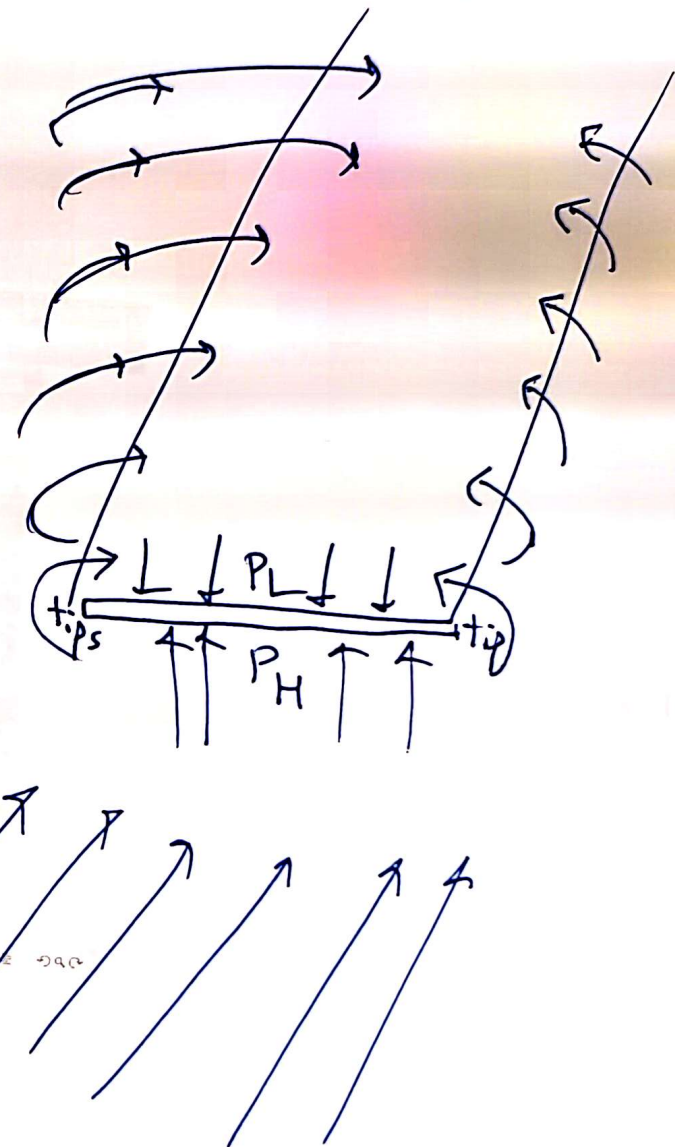
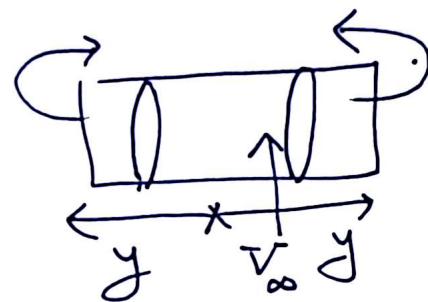


Figure 5.2: Flow over wings - qualitative effects, Fig. 7.2 from AE-BC
 AE 319, EPG Pot. Wings, Qual. Effects, 197

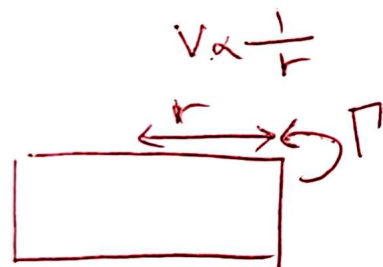


$$\vec{V} = \frac{\Gamma}{2\pi r}$$



$W \rightarrow$ due to Vortex

Every section of wing experiences two velocity $\left\{ \begin{array}{l} V_\infty \\ -W \text{ due to the Vortex} \end{array} \right.$



The induced velocity of the vortex is lower near the root due to higher r

Downwash & Induced Drag

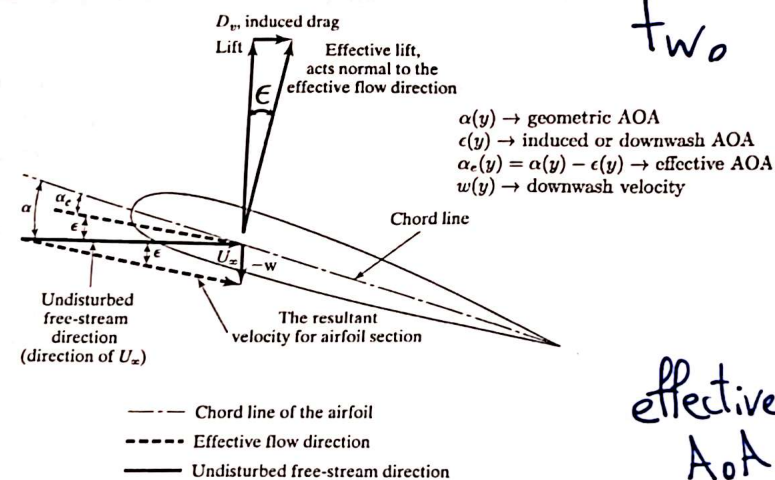
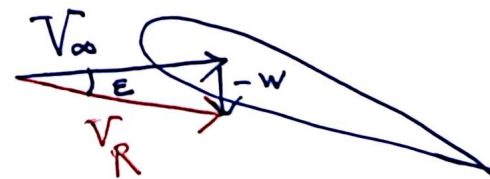


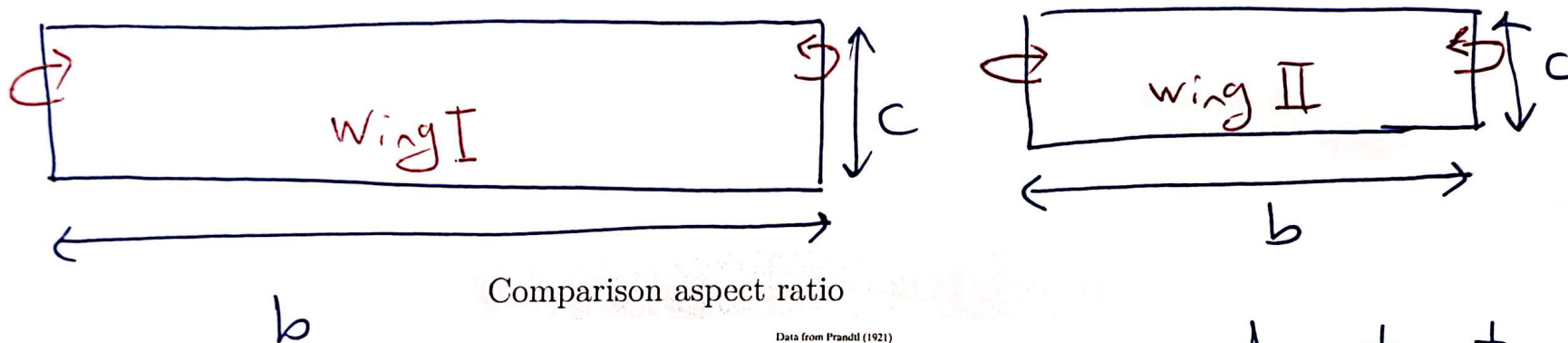
Figure 5.4: Downwash velocity and its effects, modified Fig. 7.6 from AE-BC

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effective AOA $\alpha_e = \alpha - \epsilon(y)$ geometric AOA



ϵ : downwash angle (induced angle)



Which wing has lower downwash? Wing I

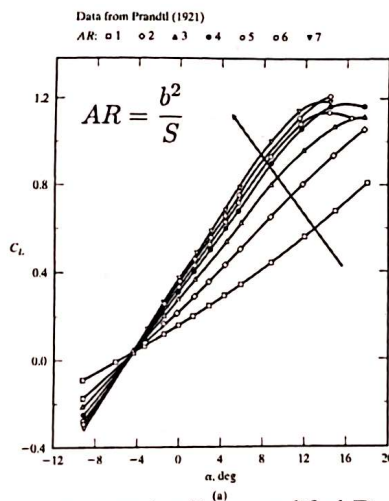


Figure 5.5: C_L vs. α aspect ratio effects, modified Fig. 7.9b from AE-BC

Note- These are from experiments and C_L is for a wing

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Aspect ratio (AR)

$$= \frac{b^2}{S} = \frac{b^2}{bc} = \frac{b}{c}$$

area for rectangular